TYPES OF GEARS





Gears - It's Classification

- Spur Gear
- Helical Gear
- Herringbone Gear
- Bevel Gear
- Worm Gear
- Rack and Pinion
- Internal and External Gear
- Face Gear
- Sprcokets

















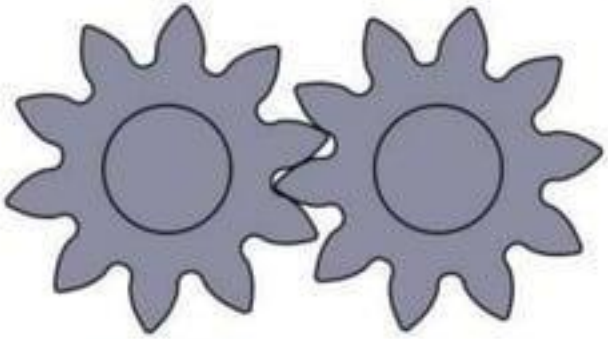


Spur Gear





- Parallel and co-planer shafts connected by gears are called spur gears.
- Spur gears have straight teeth and are parallel to the axis of the wheel.
- The advantages of spur gears are their simplicity in design, economy of manufacture and maintenance, and absence of end thrust.
- Spur gears are known as slow speed gears.
- Spur gears are the most common type of gears.

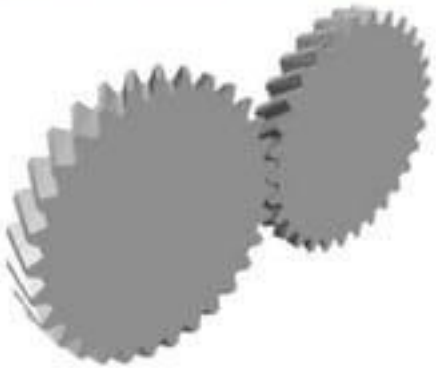




Helical Gear



- Helical gears have their teeth inclined to the axis of the shafts in the form of a helix.
- Helical gears can take higher loads than similarly sized spur gears. The motion of helical gears is smoother and quieter than the motion of spur gears.
- The angle of the helix on both the gear must be same in magnitude but opposite in direction, i.e., a right hand pinion meshes with a left hand gear.
- These gears are usually considered as high speed gears.





Herringbone Gear



- Herringbone gears resemble two helical gears that have been placed side by side.
- A disadvantage of the herringbone gear is that it cannot be cut by simple gear hobbing machines.
- Herringbone is usually referred to as Double helical gear but the fact is that Herringbone is type of double gear with two helical gears joined together correctly of opposite helix and they don't have any clearance between them, whereas in double helical gear there is a groove between the two helical gear given for tool run out for manufacturing.





Bevel/Miter Gear



Intersecting but coplanar shafts connected by gears are called bevel gears. Straight bevel gears can be used on shafts at any angle, but right angle is the most common.

Spiral Bevel Gears

- In these Spiral Bevel gears, the teeth are oblique. Spiral Bevel gears are quieter and can take up more load as compared to straight bevel gears.



Zero Bevel Gears

- Zero Bevel gears are similar to straight bevel gears, but their teeth are curved lengthwise. These curved teeth of zero bevel gears are arranged in a manner that the effective spiral angle is zero.



Worm Gear

- Worm gears are used to transmit power at 90° and where high reductions are required.
- In worm gears, one gear has screw threads. Due to this, worm gears are quiet, vibration free and give a smooth output.
- Worm gears and worm gear shafts are almost invariably at right angles.





Rack and Pinion

- A rack is a toothed bar or rod that can be thought of as a sector gear with an infinitely large radius of curvature. Pinion can be considered as a spur gear.
- Torque can be converted to linear force by meshing a rack with a pinion: the pinion turns; the rack moves in a straight line.
- Such a mechanism is used in automobiles to convert the rotation of the steeing wheel into the left-to-right motion of the tie rod(s).





Internal and External Gear

Internal Gear

- An internal gear is one with the teeth formed on the inner surface of a cylinder or cone.



External Gear

- An external gear is one with the teeth formed on the outer surface of a cylinder or cone.



Face Gear

- Face gears transmit power at (usually) right angles in a circular motion.
- Face gears are not very common in industrial application.



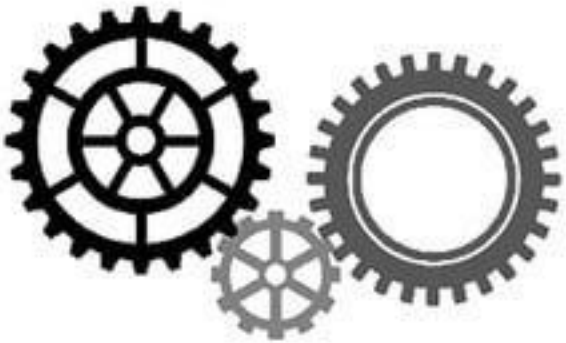


Sprockets



- Sprockets are used to run chains or belts. They are typically used in conveyor systems.
- They are used in chain drives, in motorcycles and bicycles.







Classification according to axis of shaft

- Parallel
  - Spur Gear
  - Helical Gear
  - Rack and Pinion
- Intersecting
  - Bevel Gear
- Non-intersecting and Non-parallel
  - Worm and worm gears

THANK YOU



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